

Case Study: Online Movie Ticket Booking System

In fulfillment of one of the assignments for the Information System Analysis and Design course,
taught

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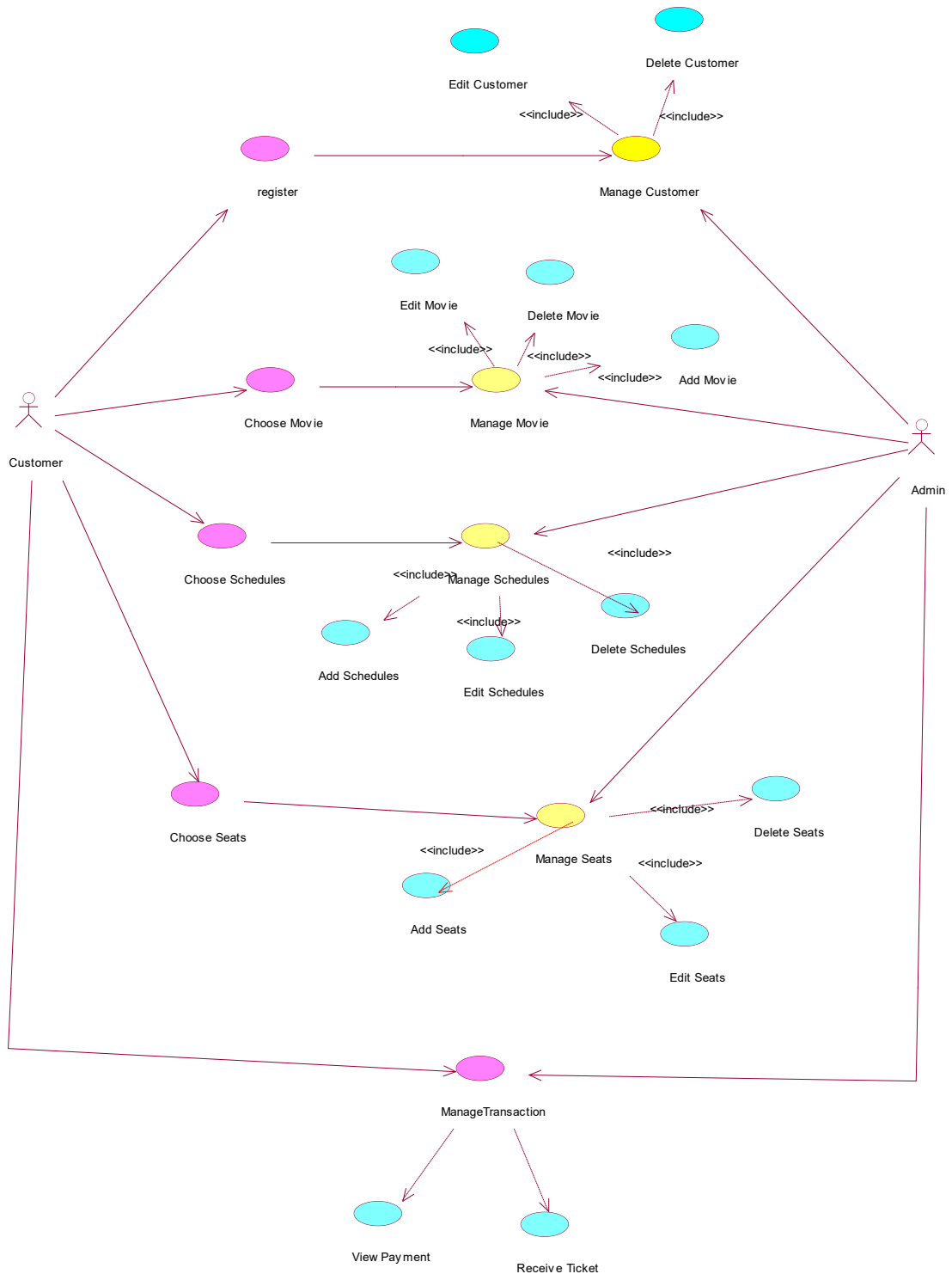
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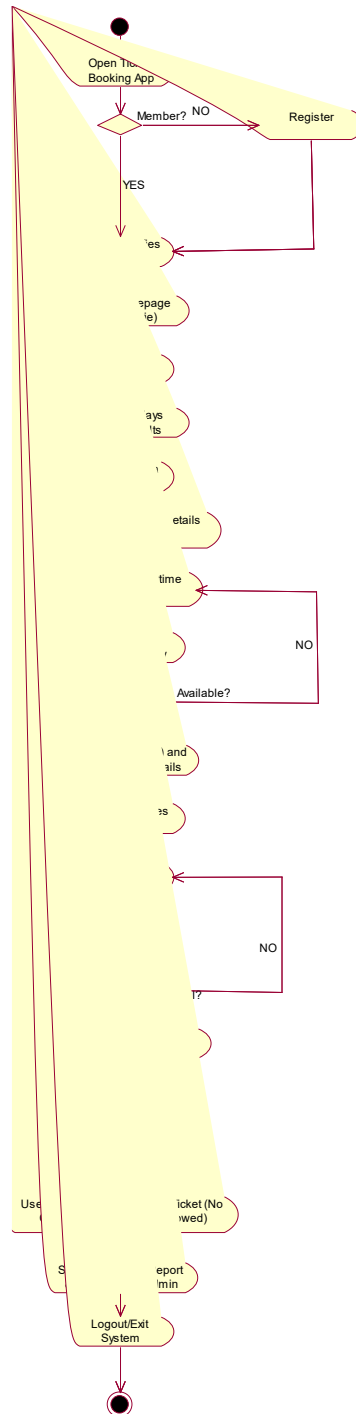
- **USE CASE DIAGRAM**



- **Actor Interaction:** Illustrates two primary actors, namely Customer and Admin.
- **Customer Functionality:** Can register, choose movies, choose schedules, choose seats, and manage transactions (view payments and receive tickets).

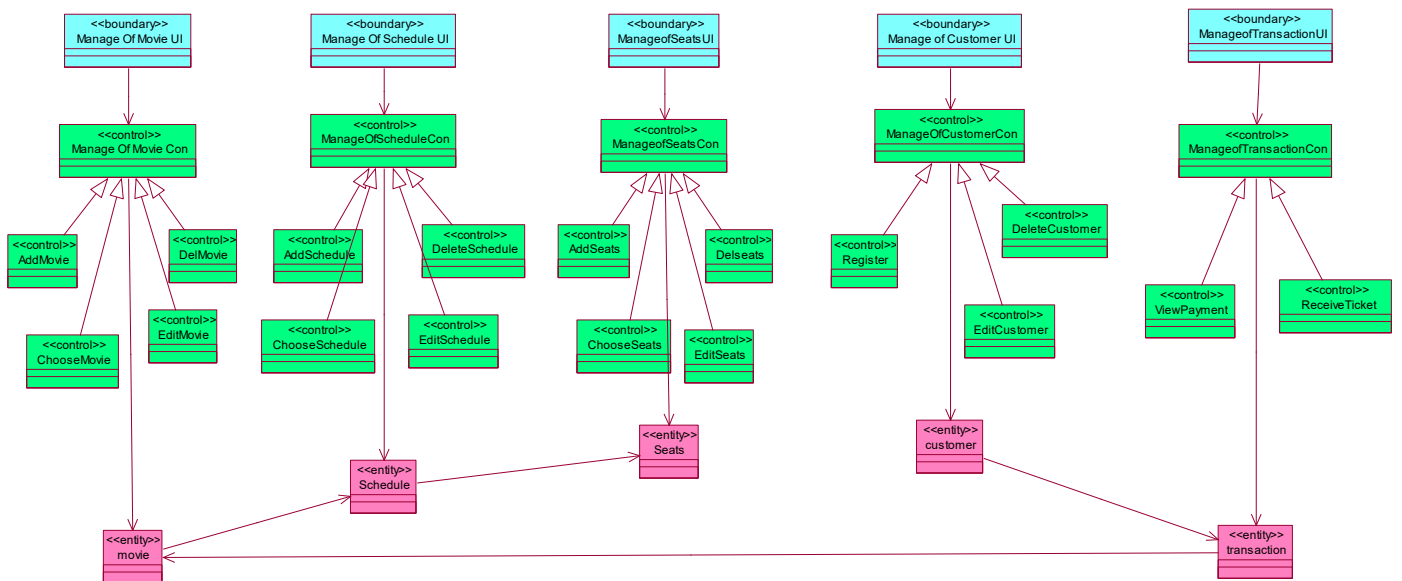
- Admin Functionality: Has full access privileges to manage master data (Manage Customer, Manage Movie, Manage Schedules, Manage Seats, and Manage Transaction) utilizing <<include>> relationships for add, update, and delete operations.

- **ACTIVITY DIAGRAM**



- **Business Process Flow:** Maps out the step-by-step workflow of the system starting from when a user opens the ticket booking application.
- **Membership Branching:** The system checks the user's membership status; if unregistered, the user is directed to complete the register process first before proceeding to select schedules, movies, seats, payment processing, and ticket generation

- **CLASS DIAGRAM**

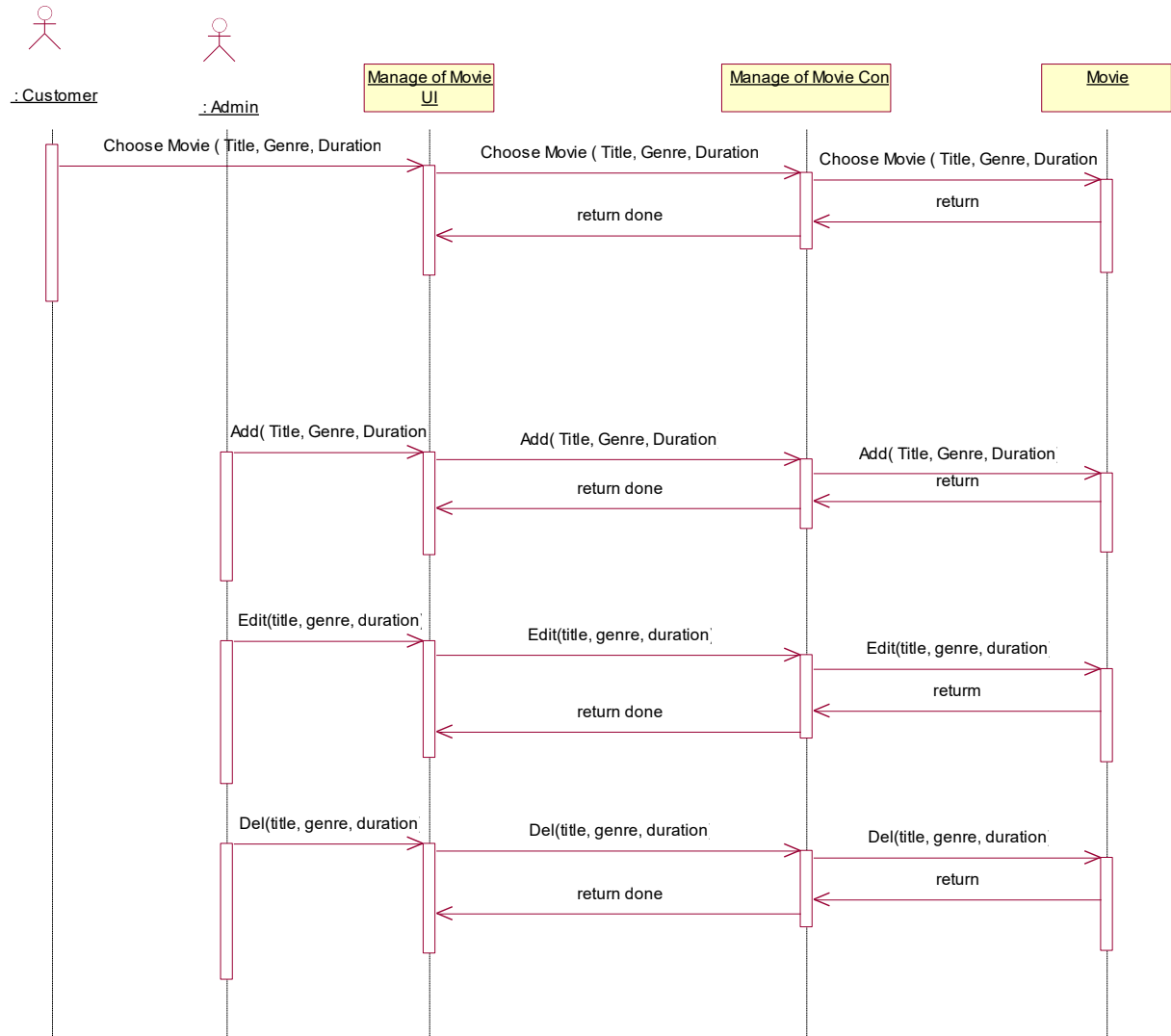


Object Structure & Relationships: Divides the architecture into standard OOP (Object-Oriented Programming) tiers: Boundary (user interface), Control (application logic), and Entity (data model).

Core Entities: Comprises classes such as movie, schedule, seats, customer, and transaction which interrelate to execute system functionalities seamlessly.

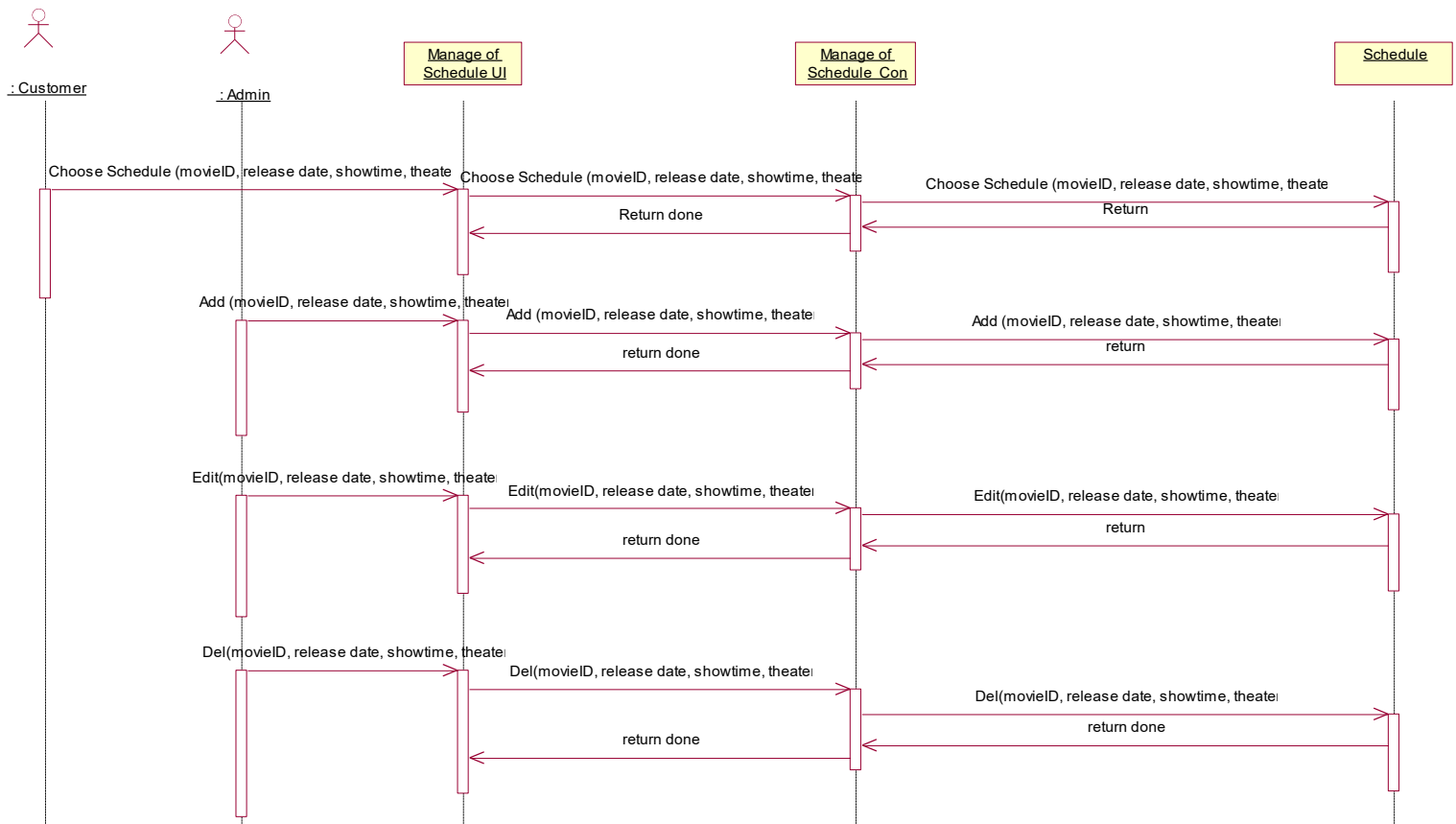
- **SEQUENCE DIAGRAM**

1. Manage of Movie



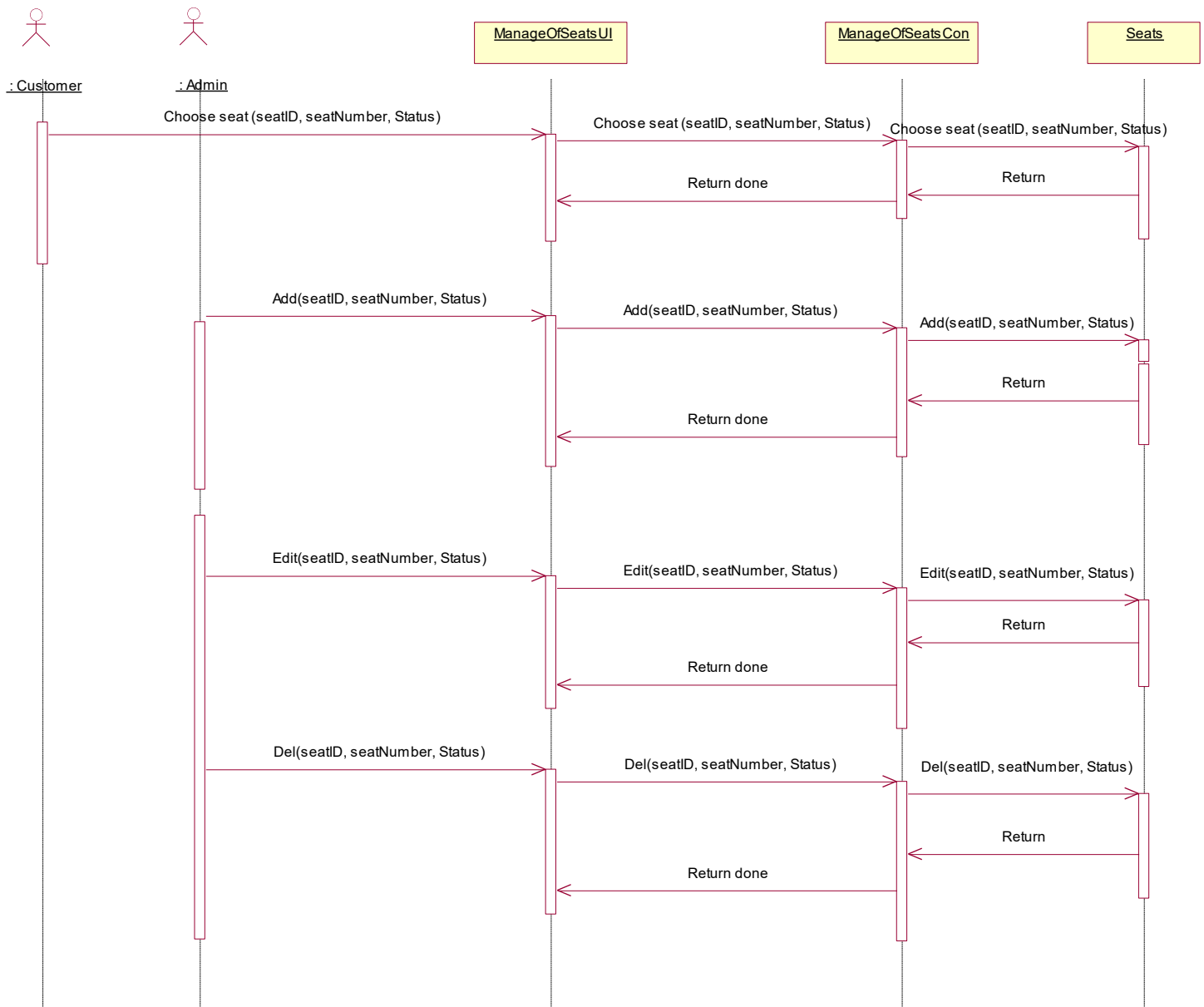
- **Manage of Movie:** Depicts message flows between Admin, UI, Controller, and the Movie entity during Choose, Add, Edit, and Delete operations.

2. Manage of Schedule



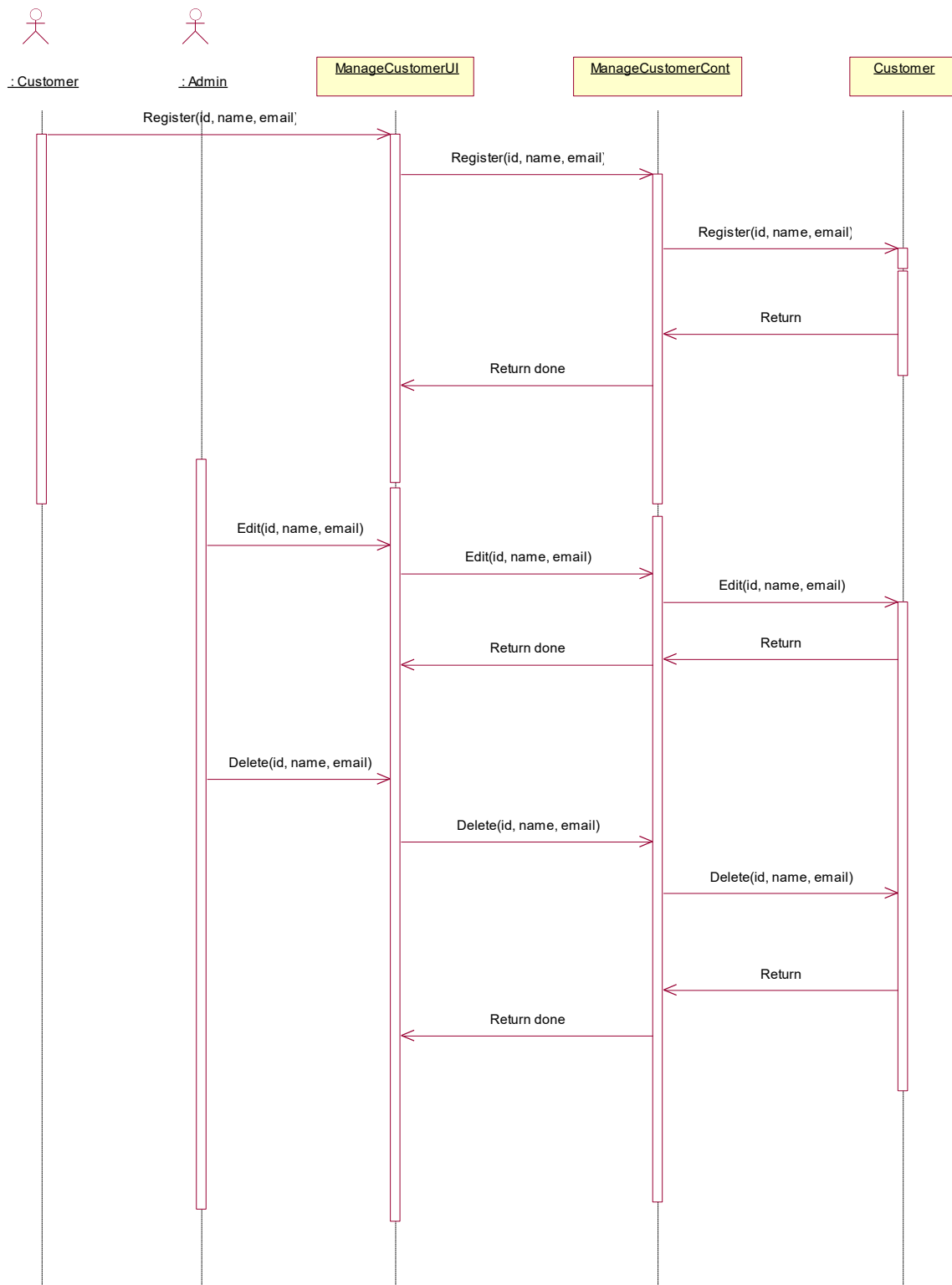
- Manage of Schedule: Synchronization flow for movie screening schedules based on Movie ID, release date, showtime, and theater.

3. Manage of Seats



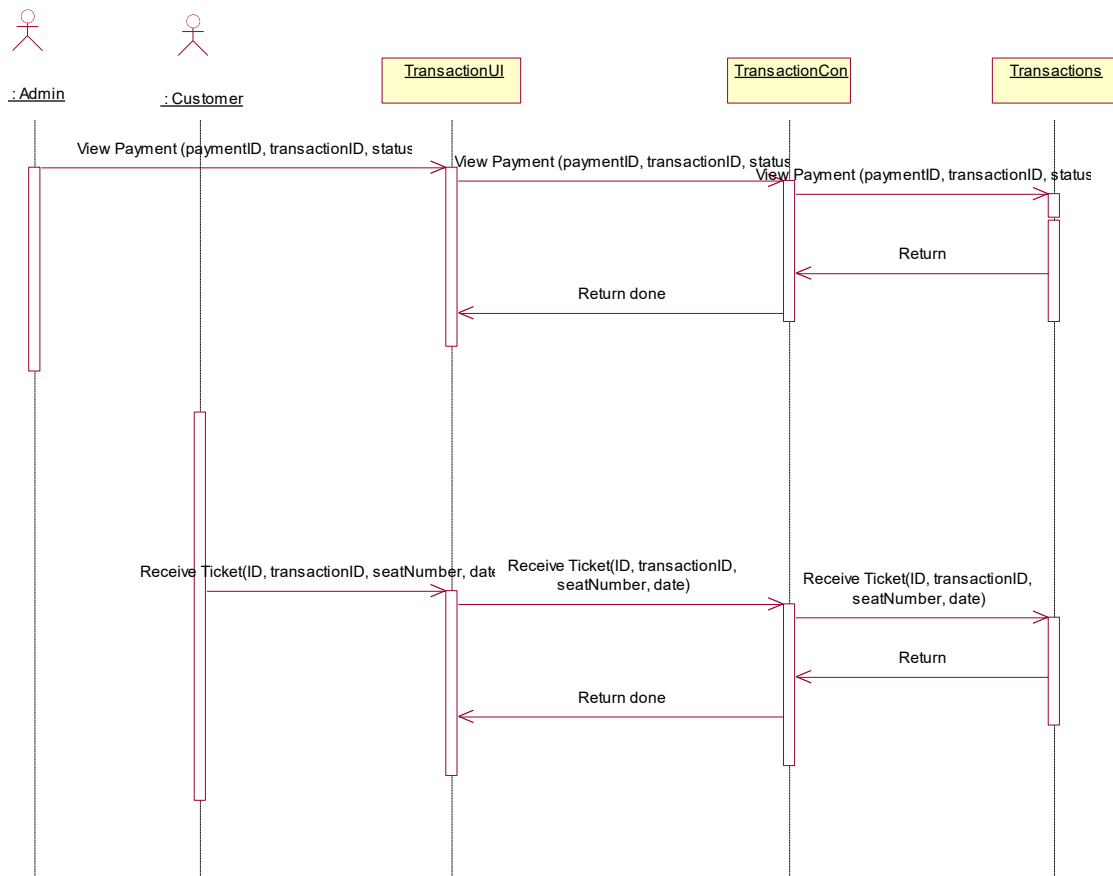
- Manage of Seats: Flow for determining and updating theater seat statuses (seatID, seatNumber, Status)

4. Manage of Customer



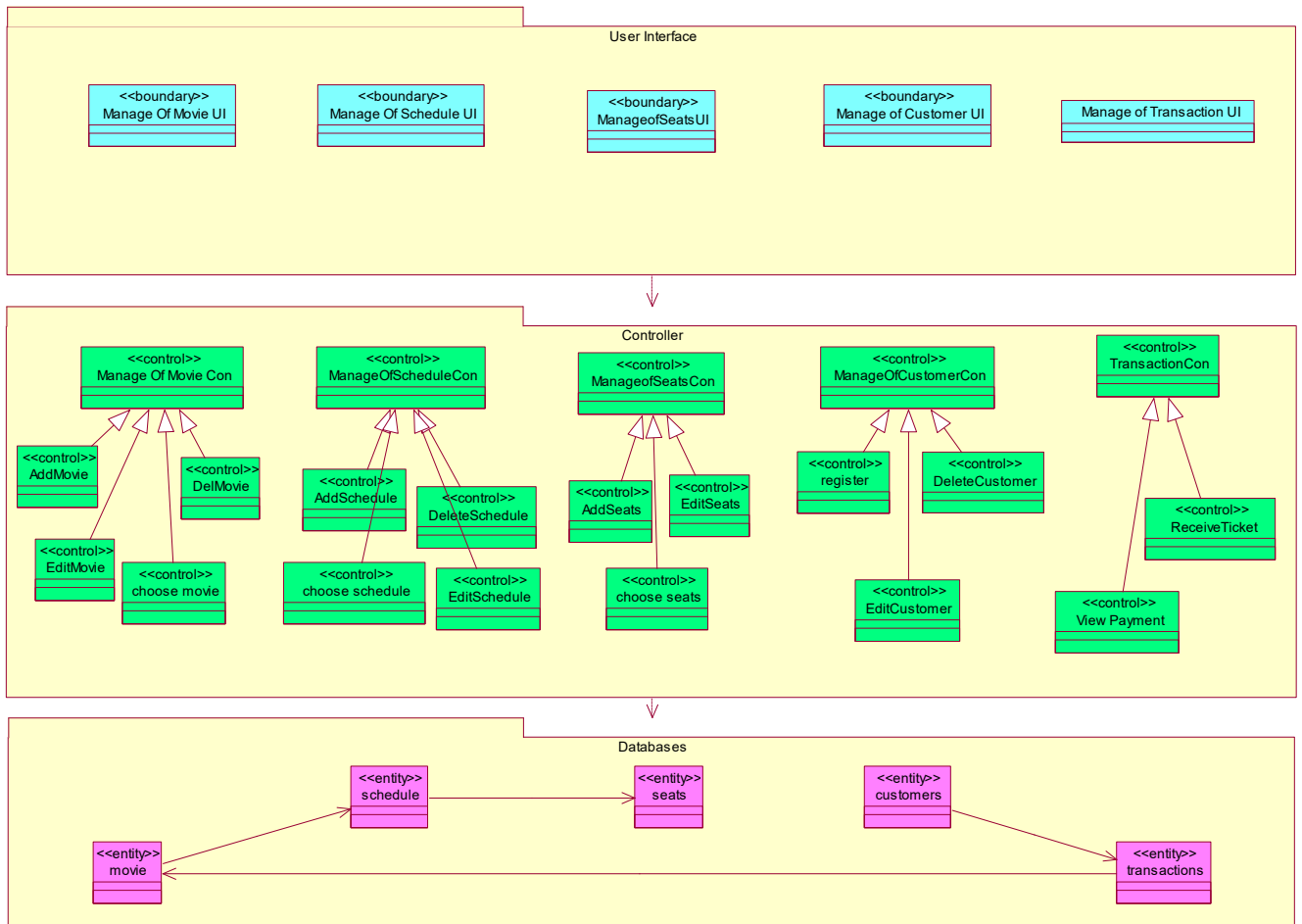
- Manage of Customer: Flow covering customer registration, profile updates, and data deletion.

5. Manage of Transaction



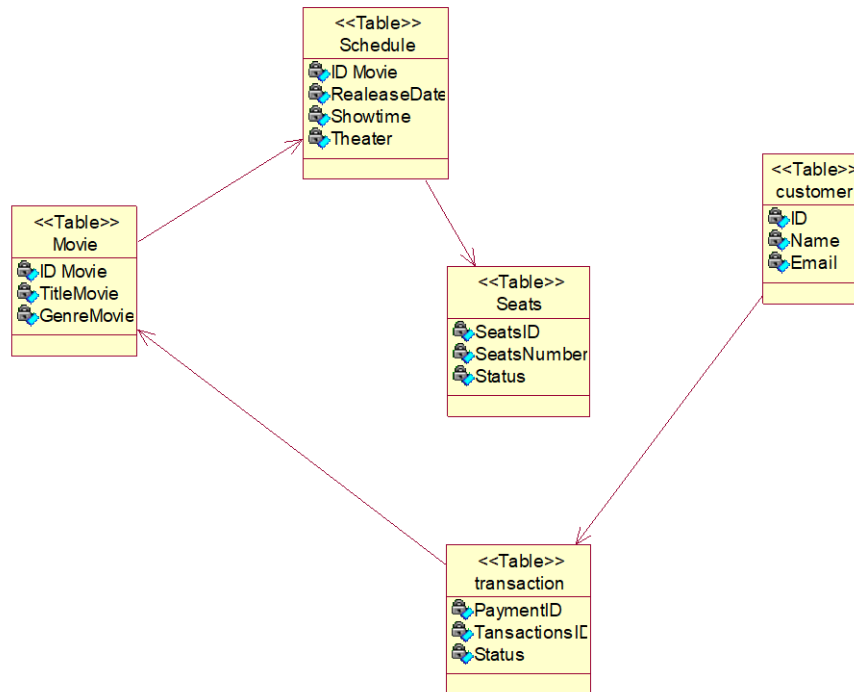
- Manage of Transaction: Communication flow to validate payment status (View Payment) and issue tickets (Receive Ticket) for customers.

6. SOFTWARE ARCHITECTURE



- Layered Architecture: Systematically organizes system components into three core tiers; User Interface (Boundary), Controller, and Databases (Entity) to ensure robust code modularity.

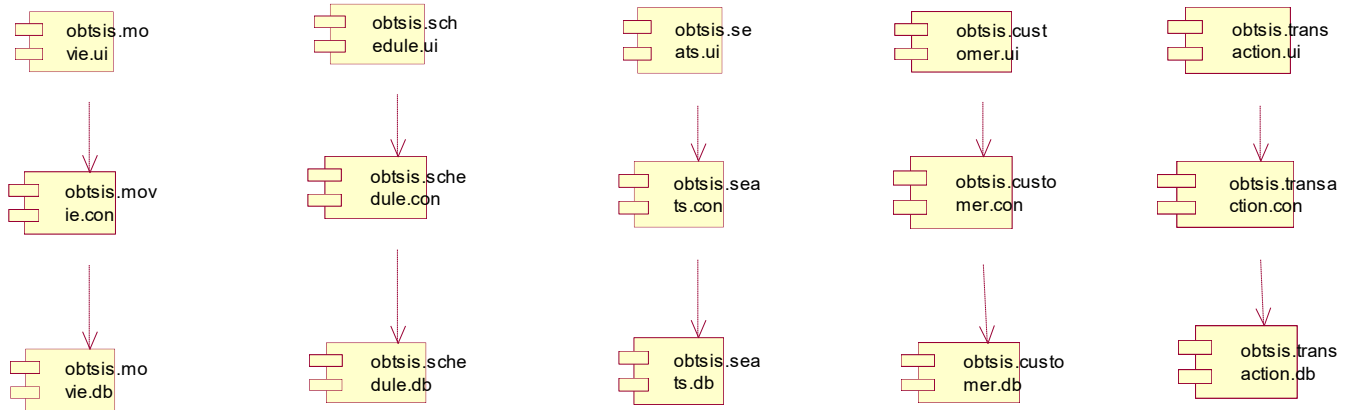
7. DATA DESIGN



Relational Database Schema: Outlines conceptual database tables along with their key attributes:

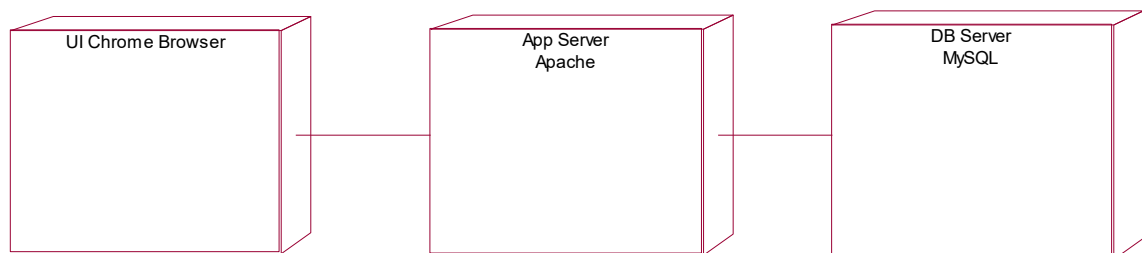
- Movie table (ID Movie, Title Movie, Genre Movie).
- Schedule table linked to movies (ID Movie, Release Date, Showtime, Theater).
- Seats table (SeatsID, Seats Number, Status).
- Customer table (ID, Name, Email).
- Transaction table (PaymentID, Transactions ID, Status).

8. COMPONENT VIEW



- **Component Modularity:** Breaks down the system into independent component modules ending in .ui, .con (controller), and .db across respective domains (Movie, Schedule, Seats, Customer, and Transaction).

9. DEPLOYMENT VIEW



Physical Infrastructure Architecture: Displays the physical deployment topology of the system following a 3-Tier Architecture model:

- **Client Tier:** UI Chrome Browser serving as the user interface.
- **Application Tier:** App Server Apache for executing application logic.
- **Data Tier:** DB Server MySQL acting as the central database repository.

10. INTERFACE DESIGN

Movie

Movie ID:

Title:

Genre:

Schedule

Movie ID:

Release date:

Showtime:

Theater:

Seats

Seats ID:

Seats Number:

Status:

Customer

ID:

Name:

Email:

Transaction

Payment ID:

Transactions ID:

Status:

- User Interface Prototypes: Simple input form wireframes for admins and the system to visually manage data entities.
- Design Consistency: Each form (Movie, Schedule, Seats, Customer, Transaction) is equipped with relevant input fields and standard action buttons (Choose, Add, Edit, Delete, Register, View Payment, Receive Ticket) for user-friendly operation.